

Final - Model Card

Basic Details

- **Model Name:** Gaussian Process Black-Box Optimizer (GP-BBO)
- **Version:** 2.0 (Post-Week 9 Global Re-aggregation Update)
- **Description:** A custom sequential optimization loop built to navigate unknown, continuous multi-dimensional landscapes without gradient information. It utilises a Gaussian Process to construct a probability distribution over the target space and uses an Expected Improvement loop to recommend the next best coordinate configurations.
- **Developer Contacts:** Student Machine Learning Developer (University Repository Maintainer)
- **Licensing:** Open Educational Use License. The underlying baseline arrays remain subject to institutional academic integrity and non-distribution guidelines.

Model Description

Input:

The model takes in a list of continuous numerical settings (X), with the number of inputs changing depending on the function space (from 2D all the way up to 8D). Every single coordinate is a raw floating-point number scaled and locked inside a standard bounding box between `0.0` and `1.0`.

Output:

The output is a single predicted score (Y) that tells us how well those specific settings performed. Because the eight functions represent completely different real-world problems, the output scales are all over the place, ranging from microscopic decimals near 10^{-124} for Function 1 to large positive numbers past `1200` for Function 5.

Model Architecture:

I built this optimization pipeline using a **Gaussian Process Regressor (GPR)** as the core statistical engine to map out the hidden surfaces.

- **The Kernel:** I chose a Matérn kernel with a smoothness parameter of $\nu = 2.5$. I bypassed the standard RBF kernel because it assumes a perfectly smooth landscape; the Matérn kernel gives the model the structural ruggedness it needs to handle sudden, sharp local peaks without over-smoothing them.
- **The Scaler:** I integrated an automated target normalisation layer (`normalize_y=True`) directly into the training loop to safely handle the mismatched output scales.

- **The Acquisition Engine:** I hand-coded an Expected Improvement (EI) loop. To find the next best point to query each week, the code maximises this EI surface globally using the L-BFGS-B algorithm, triggered from 50 completely randomised starting seeds scattered across the space to avoid local traps.

Intended Uses and Limitations

Applications Designed For:

- **Derivative-Free Global Optimization:** Perfect for tuning continuous systems where you cannot calculate derivatives or gradients (e.g., hyperparameter tuning loops, laboratory chemical recipe formulation sweeps, and complex physical simulation layout designs).
- **Sample-Efficient Tuning:** Specifically designed for scenarios where evaluating the true system is incredibly slow, expensive, or resource-constrained, making random guessing or grid search impossible.

Scenarios to Avoid:

- **Discrete or Categorical Parameters:** The model cannot naturally process categorical strings (like choosing between "A" or "B") or integer-only jumps without mapping adjustments.
- **Dynamic or Shifting Topographies:** Avoid situations involving concept drift or time-varying targets, as the model assumes the hidden landscape remains completely static.
- **Extrapolation Beyond Bounds:** Do not use this model to predict or optimize values outside the explicit `[0.0, 1.0]` boundary box, as Gaussian Process variance expands rapidly to default priors when moving outside known limits.

Training Data

Datasets and Sources:

The model was developed and trained by blending two primary data streams:

1. **Baseline Synthetic Simulator Matrices:** Pre-seeded arrays (`initial_inputs.npy` and `initial_outputs.npy`) provided by the university course instruction team. These inputs represent initial exploration points generated across eight distinct simulated domains (including compound mixing, warehouse layout strategies, and machine learning validation scores).
2. **Sequential Active Queries:** A proprietary feedback dataset compiled over a 13-week exploration horizon, where each new entry was generated by the model's previous recommendations and evaluated by the course testing API.

Evaluation Metrics

Performance is measured by tracking optimisation convergence—specifically plotting how the maximum achieved real-world score (Ymax) climbed over the 13-week search horizon. The model was trained and evaluated on the fully combined data matrix.

By the time I hit Week 13, the model locked in these peak performance scores:

[Function 1 (2D)] Max Y: 1.088195e-07		[Function 2 (2D)] Max Y: 0.737285
[Function 3 (3D)] Max Y: -0.023599		[Function 4 (4D)] Max Y: -3.820595
[Function 5 (4D)] Max Y: 1258.716961		[Function 6 (5D)] Max Y: -0.636036
[Function 7 (6D)] Max Y: 1.364970		[Function 8 (8D)] Max Y: 9.598482

Fairness Assessments:

Because this data is entirely abstract, synthetic, and mathematical, it does not involve human subjects, demographic attributes, or social classification tasks. Traditional algorithmic fairness categories (such as gender, race, or age-group parity) do not apply here.

Instead, "fairness" was evaluated as uniform algorithmic coverage across the different function granularities. Early on, the model performed poorly on high-dimensional spaces compared to low-dimensional spaces due to sample sparsity. The Week 9 re-aggregation update corrected this system-wide performance imbalance, allowing the model to converge optimally regardless of input dimensionality (2D vs. 8D).

Trade-offs

- **The Short-Term Leaderboard Trap:** Keeping the acquisition jitter (ξ) low (0.01) guarantees safe, incremental weekly improvements to protect your ranking, but it creates a massive blind spot. This safe bias blindfolded my model, preventing it from looking for massive peaks further away and causing my 6D and 8D scores to stall out completely mid-project.
- **Compute Time vs. Breaking Local Traps:** Cranking the L-BFGS-B acquisition optimizer up to 50 random global restarts successfully breaks out of deceptive local dead ends, but it makes the script take significantly longer to compute and suggest the next recommended vector.
- **Extreme Selection Bias:** Because the data was gathered by an active optimization algorithm, the points are heavily clustered right around the highest peaks. This makes the model incredibly precise at navigating the global maxima, but it leaves it with massive variance and high prediction errors in the low-performing valleys where it purposefully chose not to sample.

Ethical Considerations

- **Identified Bias (Selection Bias):** The model is mathematically biased toward its own historical successes. Because Bayesian Optimization inherently seeks to maximize scores, the model actively starves low-performing regions of attention. This creates a deeply skewed dataset that cannot be safely repurposed to train standard, downstream regression networks without introducing massive sampling errors.
- **Risks of Stagnation:** In high-stakes real-world settings (like health-tech hyperparameter tuning or structural engineering checks), setting exploration parameters too low carries the financial and operational risk of permanently locking into an inferior local maximum while completely missing the true global peak.
- **Risk Mitigation Strategies:** To counter lengthscales collapse and algorithmic myopia, the framework integrates a dynamic global restart system (50 random multi-start optimizer runs) and forces full baseline history re-aggregation. If deployed in real-world environments, developers should implement a random epsilon-greedy query step (e.g., forcing a purely random exploratory point 5–10% of the time) to keep the Gaussian Process from developing structural blind spots.